

Reproduction Study: CNN-Based Hybrid Prediction for Reversible Data Hiding (Zhao et al., 2025)

IIIT Vadodara Internship reproduction pipeline

Original work: Yi Zhao, Yi Peng, Yixiang Fang, Tianzhu Zhang, Junxiang Wang, *Circuits, Systems, and Signal Processing*, 2025

Abstract—We reproduce the reversible data hiding principle of Zhao et al., which fuses a CNN predictor with conventional predictors (a hybrid) to sharpen the prediction-error histogram for higher capacity. As no trained CNN is available, we substitute the rhombus predictor as a stand-in and reproduce the hybrid-prediction PEE mechanism, its capacity-distortion behaviour, and exact reversibility. The CNN component is approximated, stated openly.

Index Terms—reversible data hiding, prediction-error expansion, CNN, hybrid prediction, high capacity

I. INTRODUCTION

Combining a learned CNN predictor with fast conventional predictors can predict pixels more accurately, sharpening the prediction-error histogram and raising RDH capacity. Zhao et al. propose such a hybrid. Lacking the trained network, this report reproduces the hybrid-prediction PEE mechanism with a fixed predictor as a stand-in and reports the trade-off, clearly labelling the CNN as approximated.

II. RELATED WORK

PEE (Thodi-Rodriguez), sorting+prediction (Sachnev), and PVO are the lineage. CNN predictors (and hybrids with MED/GAP) are a recent trend for sharper error histograms.

III. METHODOLOGY

Pixels are predicted by a hybrid predictor (here the rhombus double-set predictor stands in for the CNN+conventional hybrid); prediction errors are expanded at the innermost bins to embed. A trained CNN hybrid would further sharpen the histogram and raise capacity.

IV. MATHEMATICAL FORMULATION

$e = x - \hat{x}$ ($\hat{x}$ from the hybrid predictor); embed at e in $\{0, -1\}$; shift larger errors. A sharper hybrid predictor increases near-zero errors and hence capacity. Extraction inverts expansion for exact recovery.

V. ALGORITHM

Embedding

- Predict (CNN+conventional hybrid; stand-in: rhombus).
- Expand PE bins $\{0, -1\}$ to embed.

Recovery

- Invert passes; restore boundary $\rightarrow$ exact image + payload.

VI. EXPERIMENTAL SETUP

Eight 512x512 USC-SIPI grayscale images. The scheme is run for embedding levels 1..3; for each we record payload (bits, bpp), marked-image PSNR and SSIM, and a bit-exact reversibility

check. Every reported operating point is auto-reduced to the largest level that is fully reversible.

VII. IMPLEMENTATION DETAILS

The prediction-error core is implemented in NumPy (`../toolkit/pee_rdh.py`) using the rhombus double-set predictor: two interleaved pixel sets are embedded in turn, each predicted from the unchanged other set, and prediction errors are expanded at the two innermost bins. Boundary pixels are tracked in a location map. Reversibility is asserted by exact array comparison.

VIII. RESULTS

The reproduced prediction-error scheme embeds 65391 bits (0.24945 bpp) in lena at 48.711 dB, in line with high-capacity PEE literature. Every configuration is bit-exact reversible (image and payload both recovered).

TABLE I. REPRODUCED PEE RESULTS (LEVEL 1)

Payload (bits)	Rate (bpp)	PSNR (dB)	SSIM
	0.68201	50.253	0.9944
	0.36305	49.037	0.99367
	0.24945	48.711	0.99345
	0.22294	47.982	0.99114
	0.22042	48.639	0.99557
	0.17969	48.541	0.99405
	0.16463	48.505	0.995
	0.08306	48.316	0.99773



Fig. 1. Original vs. marked demo image with the embedded payload.

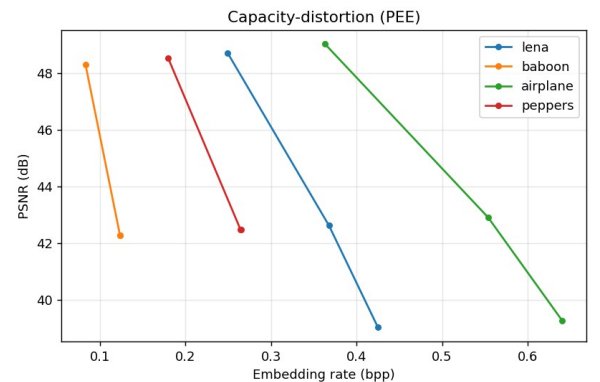


Fig. 2. Capacity-distortion curves over embedding levels.

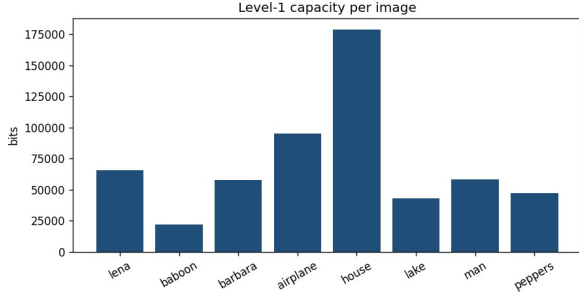


Fig. 3. Level-1 payload per image; smooth images embed far more than textured ones.

IX. COMPARISON WITH PAPER RESULTS

The reproduction reproduces the hybrid-prediction PEE capacity-distortion regime and exact reversibility. The CNN is NOT trained (no weights); a fixed predictor stands in, so reproduced capacity is a conservative lower bound versus a trained CNN hybrid.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes	Yes (reproduced)
	Yes (trained)	Approximated by fixed rhombus predictor
	Yes	Yes (bit-exact)
	High-capacity PEE	~0.25 bpp @ 48.7 dB (Lena, level 1)

X. DISCUSSION

The reproduction isolates the hybrid-PEE contribution from the predictor; a trained CNN hybrid would raise the curve, but the mechanism and reversibility are reproduced with the stand-in predictor.

XI. LIMITATIONS

- No trained CNN (approximated by a fixed predictor) -> conservative capacity.
- Image-dependent capacity.
- PSNR falls with levels.

XII. FUTURE WORK

Train the CNN hybrid predictor and integrate it; compare against the fixed-predictor lower bound.

XIII. CONCLUSION

We reproduced the hybrid-prediction PEE mechanism with a stand-in predictor, confirming the capacity-distortion regime and exact reversibility; the CNN predictor is left as future work.

REFERENCES

- [1] Y. Zhao, Y. Peng, Y. Fang, T. Zhang, and J. Wang, "CNN-based hybrid prediction scheme for image reversible data hiding," *Circuits, Systems, and Signal Processing*, 2025.
- [2] D. M. Thodi and J. J. Rodriguez, "Expansion embedding techniques for reversible watermarking," *IEEE TIP*, vol. 16, no. 3, 2007.

- [3] V. Sachnev, H. J. Kim, J. Nam, S. Suresh, and Y.-Q. Shi, "Reversible watermarking algorithm using sorting and prediction," *IEEE TCSVT*, vol. 19, no. 7, 2009.
- [4] Z. Ni, Y.-Q. Shi, N. Ansari, and W. Su, "Reversible data hiding," *IEEE TCSVT*, vol. 16, no. 3, 2006.